Privacy-preserving Genome Path Release against a Public Pangenome Graph

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Abstract

Human pangenome graphs represent any genome as a path through shared coordinates, but releasing that path risks re-identification. Existing methods do not release paths for out-of-graph genomes and do not provide a stated bound on identification. We introduce PathGuard, which implements a randomized path release mechanism whose output is sampled from a distribution over cohort-supported paths in the public pangenome graph. Every release carries a privacy level set in advance with a guaranteed bound on how accurately an equal-prior attacker can distinguish the target's release from a decoy. PathGuard builds a Li-Stephens model over the public cohorts haplotypes to define a baseline distribution over cohort-supported paths. This baseline is exponentially tilted towards paths agreeing with the target, and releases are drawn by forward-filtering, backward-sampling. We prove that the release distributions induced by any two input paths have a bounded total variation distance, and thus establish a formal upper bound on their distinguishability by any equal-prior adversary, even one with full knowledge of the pangenome graph, release mechanism, target genome, and decoy genome. Benchmarked against PanMixer for cohort path obfuscation, our path-release strategy resists its published linkage, membership-inference, and reconstruction attacks while altering only 17.5% of the target path's cells (82.5% of target cells retained), a level of distortion comparable to that of PanMixer at its most privacy-protective setting. Read-mapping rates stay within 0.063 points of the unobfuscated baseline and every release still carries a privacy ceiling. By bounding the identification risk associated with a released path below a predefined threshold, PathGuard provides a general framework for sharing the pangenome path of any individual genome while offering a quantifiable upper bound on the information disclosed.
Keywords: Pangenome graph; privacy-preserving path release; backward sampling; randomized algorithm

1 Introduction

Pangenome graphs represent genomic variation using paths through a shared sequence graph and support applications including read alignment, genotyping, and variant calling [2]. Once a graph has been constructed and released, a new genome can be mapped or projected onto the graph. The resulting path may itself be privacy-sensitive, even when the graph and its construction cohort are public. In this paper, we aim to develop a randomized genome path release mechanism with the theoretical guarantee that the output distributions induced by any two input paths can be distinguished with the probability of no more than a specified pre-specified privacy parameter

$\tau \in [0, 1)$. Specifically, we consider as the input a fixed pangenome graph G constructed from a public cohort $D = \{g_1, \ldots, g_n\}$ and an external target genome $g \notin D$ and its mapped path p_g through G, and our objective is to release a randomized sanitized path while limiting the statistical distinguishability of the target from individuals represented in D.

The proposed mechanism uses a target-independent cohort path model as a baseline distribution. It then introduces a bounded preference for paths similar to the target. The preference is calibrated so that it cannot make output distributions induced by different input paths differ by more than a prescribed total variation distance.

2 Methods

2.1 Fixed graph and mapped paths

Let G be a fixed pangenome graph constructed from the cohort $D = \{g_1, \ldots, g_n\}$. For each cohort genome g_i, let $p_i = \mathrm{Map}(g_i, G)$ be its graph path. Let $g \notin D$ be an external target genome and $p_g = \mathrm{Map}(g, G)$. The graph G, the cohort D, the cohort paths $\{p_i\}_{i=1}^n$, and the release algorithm are assumed to be known to the attacker. The only private input is the target path supplied to the release mechanism.

Let $\mathcal{P}(G)$ denote the admissible input-path space. Let $\mathcal{Y}_D \subseteq \mathcal{P}(G)$ denote a fixed set of cohort-supported output paths. The set $\mathcal{Y}_D$ may contain the complete cohort haplotype paths, the synthetic paths generated by a cohort haplotype model, the mosaics of cohort-supported local graph traversals or graph-valid paths generated by an HMM model. The output support $\mathcal{Y}_D$ must be fixed independently of the target path.

Let $M_{\mathrm{path}}(p; G, D)$ be a randomized path release mechanism. For an input path p, we define $Q_p = \mathrm{Law}(M_{\mathrm{path}}(p; G, D))$. Thus, Q_p is the distribution of sanitized paths released when the input is p. For two probability distributions P and Q on $\mathcal{Y}_D$, we define total variation distance as $\mathrm{TV}(P, Q) = \sup_{A \subseteq \mathcal{Y}_D} |P(A) - Q(A)|$. When $\mathcal{Y}_D$ is finite,

$$\mathrm{TV}(P, Q) = \frac{1}{2} \sum_{y \in \mathcal{Y}_D} |P(y) - Q(y)|.$$

Definition 1 (τ-private path release against a cohort)**.** *For a target path p_g and public cohort paths $p_1, \ldots, p_n$, a mechanism M_{path} is τ-private for p_g against D if* $\max_{1 \le i \le n} \mathrm{TV}(Q_{p_g}, Q_{p_i}) \le \tau$. *Alternatively, a stronger condition is* $\sup_{p,q \in \mathcal{P}(G)} \mathrm{TV}(Q_p, Q_q) \le \tau$.

Under equal prior probabilities for the hypotheses: $H_0 : \widetilde{P} \sim Q_p$ and $H_1 : \widetilde{P} \sim Q_q$, where $\widetilde{P}$ is the released path, the optimal binary testing success probability (i.e., to distinguish any two paths p and q) is $P^*_{\mathrm{succ}}(p, q) = \frac{1+\mathrm{TV}(Q_p, Q_q)}{2}$. Therefore, $\mathrm{TV}(Q_p, Q_q) \le \tau$ implies $P^*_{\mathrm{succ}}(p, q) \le \frac{1+\tau}{2}$. For example, $\tau = 0.10$ bounds optimal equal-prior binary identification accuracy by 0.55.

2.2 Cohort-Tilted τ-Private Path Release

The mechanism proposed below satisfies this stronger pairwise condition, which guarantees the mechanism is private for any input path.

Let R_D be a probability distribution on $\mathcal{Y}_D$ constructed only from the public graph and public cohort. We assume $R_D(y) > 0$ for every $y \in \mathcal{Y}_D$, and both the support $\mathcal{Y}_D$ and the baseline probabilities $R_D(y)$ are independent of the private input path.

Let $u : \mathcal{P}(G) \times \mathcal{Y}_D \to [0, 1]$ be a bounded utility function so that a larger value of $u(p, y)$ means that releasing y preserves more utility for input path p. A useful blockwise construction of u is

$u(p, y) = \sum_{t=1}^{T} \beta_t \phi_t(p_t, y_t)$, where $\beta_t \geq 0$, $\sum_{t=1}^{T} \beta_t = 1$, and $0 \leq \phi_t(p_t, y_t) \leq 1$. The utility ϕ_t may measure the local agreement between the alleles in the two paths p and y. For example, a population-weighted agreement can be defined on k-mer as

$$\phi_t(a, b) = \frac{\sum_x w(x) \min\{\mathbf{1}[x \in K(a)], \mathbf{1}[x \in K(b)]\}}{\sum_x w(x) \max\{\mathbf{1}[x \in K(a)], \mathbf{1}[x \in K(b)]\}},$$

where $w(x)$ downweights ubiquitous human k-mers and gives greater weight to population-informative k-mers.

For $0 \leq \tau < 1$, we define $\eta_\tau = \operatorname{arctanh}(\tau) = \frac{1}{2} \log \frac{1+\tau}{1-\tau}$. For input path p, we define the release distribution $Q_p^\tau(y) = R_D(y) \exp(\eta_\tau u(p, y)) / Z_p$, for $y \in \mathcal{Y}_D$, where Z_p is a normalization factor: $Z_p = \sum_{z \in \mathcal{Y}_D} R_D(z) \exp(\eta_\tau u(p, z))$, and the path release mechanism as: $M_\tau(p; G, D) : \widetilde{P} \sim Q_p^\tau$. The mechanism may be interpreted as the foolowing steps:

1. R_D restricts outputs to paths supported by the public cohort model;
2. $u(p, y)$ favors cohort-supported outputs that resemble the target;
3. η_τ limits how strongly the private target may change the baseline distribution.

When $\tau = 0$, $\eta_\tau = 0$, and $Q_p^0 = R_D$ for every input path p, where the output is randomly drawn from R_D and independent of the target. As τ approaches 1, the mechanism is allowed to concentrate more strongly on high-utility outputs.

2.3 Privacy Guarantee

Theorem 1 (Pairwise path release guarantee)**.** *For any two admissible input paths $p, q \in \mathcal{P}(G)$, $\mathrm{TV}(Q_p^\tau, Q_q^\tau) \leq \tau$. Moreover, the mechanism satisfies the pointwise likelihood-ratio bound*

$$e^{-\varepsilon_\tau} \leq \frac{Q_p^\tau(y)}{Q_q^\tau(y)} \leq e^{\varepsilon_\tau}, \qquad \varepsilon_\tau = 2 \operatorname{arctanh}(\tau) = \log \frac{1+\tau}{1-\tau}.$$

Proof. Because $0 \leq u(p, y) \leq 1$, we have $1 \leq \exp(\eta_\tau u(p, y)) \leq e^{\eta_\tau}$. Since R_D is a probability distribution, $1 \leq Z_p \leq e^{\eta_\tau}$ for every input path p. Consequently, $e^{-\eta_\tau} \leq Z_q / Z_p \leq e^{\eta_\tau}$. For every $y \in \mathcal{Y}_D$,

$$\frac{Q_p^\tau(y)}{Q_q^\tau(y)} = \exp(\eta_\tau \left[u(p, y) - u(q, y)\right]) \frac{Z_q}{Z_p}.$$

Since $-1 \leq u(p, y) - u(q, y) \leq 1$, we obtain $e^{-\eta_\tau} \leq \exp(\eta_\tau [u(p, y) - u(q, y)]) \leq e^{\eta_\tau}$. Combining the two bounds gives $e^{-2\eta_\tau} \leq Q_p^\tau(y) / Q_q^\tau(y) \leq e^{2\eta_\tau}$. A standard likelihood-ratio bound implies

$$\mathrm{TV}(Q_p^\tau, Q_q^\tau) \leq \frac{e^{2\eta_\tau} - 1}{e^{2\eta_\tau} + 1} = \tanh(\eta_\tau) = \tau.$$

□

Remark 1. *The likelihood-ratio statement is stronger than the requested total variation guarantee. It is an ε_τ-local differential privacy guarantee over the admissible input-path domain, with the budget $\varepsilon_\tau = \log \frac{1+\tau}{1-\tau}$. In this paper, we nevertheless uses τ because it has the direct identification interpretation $P_{\mathrm{succ}}^* \leq \frac{1+\tau}{2}$.*

2.4 Privacy relative to a cohort mixture

Let $\pi = (\pi_1, \ldots, \pi_n)$ be any public prior over cohort members, where $\pi_i \geq 0$, $\sum_i \pi_i = 1$. We define the cohort-mixture release distribution $Q_D^{\pi,\tau} = \sum_{i=1}^n \pi_i Q_{p_i}^\tau$.

Corollary 1 (Cohort-mixture guarantee)**.** *For every target path p_g,* $\mathrm{TV}\left(Q_{p_g}^\tau, Q_D^{\pi,\tau}\right) \leq \tau$.

Proof. By convexity of total variation distance,

$$\begin{aligned}\mathrm{TV}\left(Q_{p_g}^\tau, \sum_i \pi_i Q_{p_i}^\tau\right) &\leq \sum_i \pi_i \,\mathrm{TV}\left(Q_{p_g}^\tau, Q_{p_i}^\tau\right) \\ &\leq \sum_i \pi_i \tau = \tau.\end{aligned}$$

□

Thus, the release of the external target is difficult to distinguish both from every individual cohort release and from a release generated by a randomly selected cohort member.

2.5 Conditions required by the guarantee

It was worth noting that the theoretical analysis depends on three conditions:

1. **Common support:** $\mathcal{Y}_D$ must be fixed independently of the target. Target-specific candidate lists that introduce target-dependent zero probabilities generally invalidate the guarantee.
2. **Target-independent baseline:** R_D must be learned from G and D only.
3. **Globally bounded target utility:** the target-dependent utility must satisfy $0 \leq u(p, y) \leq 1$.

If a raw utility has range $u_{\max} - u_{\min} = \Delta_u$, the corresponding calibration is $\eta_\tau = \operatorname{arctanh}(\tau)/\Delta_u$.

2.6 HMM-Based Implementation

We use forward-filtering coupled with backward sampling to obtain the to-be-released path. Suppose a graph path is divided into T ordered sites or graph blocks. Let $Z_t \in \{1, \ldots, K\}$ be a hidden state indicating one of K cohort haplotypes or valid cohort-supported local graph traversals at position t. Let $h_{t,j}$denote the local graph traversal supplied by state j at position t. The target-independent cohort HMM is

$$R_D(z_{1:T}) = \rho(z_1) \prod_{t=2}^{T} A_t(z_{t-1}, z_t),$$

where:

- $\rho(j)$ is the initial-state distribution;
- $A_t(i, j)$ is the transition probability from state i to state j;
- transitions with probability zero encode graph-incompatible concatenations.

A hidden-state sequence $z_{1:T}$ induces the released graph path $\Phi(z_{1:T}) = h_{1,z_1} \| h_{2,z_2} \| \cdots \| h_{T,z_T}$. Let the external target path be $p = (p_1, \ldots, p_T)$. We choose $\phi_t(p_t, h_{t,j}) \in [0,1]$ and weights $\beta_t \geq 0$, $\sum_{t=1}^{T} \beta_t = 1$. Denoting $u(p, z_{1:T}) = \sum_{t=1}^{T} \beta_t \phi_t(p_t, h_{t,z_t})$, the tilted HMM distribution is $Q_p^\tau(z_{1:T}) \propto R_D(z_{1:T}) \exp(\eta_\tau u(p, z_{1:T}))$. Because the utility is additive,

$$Q_p^\tau(z_{1:T}) \propto \rho(z_1)\psi_1^p(z_1) \prod_{t=2}^{T} A_t(z_{t-1}, z_t)\psi_t^p(z_t)$$

, where $\psi_t^p(j) = \exp(\eta_\tau \beta_t \phi_t(p_t, h_{t,j}))$. The cohort HMM remains fixed. The private target changes only the bounded local potentials.

Forward filtering. We define an unnormalized forward value $r_t(j)$ and a normalized forward probability $\alpha_t(j)$. Initialize $r_1(j) = \rho(j)\psi_1^p(j)$, $c_1 = \sum_{j=1}^{K} r_1(j)$, and $\alpha_1(j) = r_1(j)/c_1$.

For $t = 2, \ldots, T$, we compute $r_t(j) = \psi_t^p(j) \sum_{i=1}^{K} \alpha_{t-1}(i) A_t(i,j)$, $c_t = \sum_{j=1}^{K} r_t(j)$, and $\alpha_t(j) = r_t(j)/c_t$. Per-position normalization prevents numerical underflow. The log normalizing constant is $\log Z_p = \sum_{t=1}^{T} \log c_t$.

Backward sampling. We sample the final state as $Z_T \sim \text{Categorical}(\alpha_T(1), \ldots, \alpha_T(K))$. For $t = T-1, \ldots, 1$, conditional on $Z_{t+1} = j$, sample Z_t according to

$$\Pr[Z_t = i \mid Z_{t+1} = j, p] = \frac{\alpha_t(i) A_{t+1}(i,j)}{\sum_{\ell=1}^{K} \alpha_t(\ell) A_{t+1}(\ell, j)}.$$

Finally, we release $\widetilde{p} = \Phi(Z_{1:T})$. Unlike Viterbi decoding that always samples the block with the maximum probability, the forward sampling algorithm samples from the desired randomized release distribution. Because $\widetilde{p}$ is a deterministic post-processing of $Z_{1:T}$, $\text{TV}(\text{Law}(\widetilde{p} \mid p), \text{Law}(\widetilde{p} \mid q)) \leq \text{TV}(Q_p^\tau, Q_q^\tau) \leq \tau$. The overall algorithm is summarized in the Algorithm below.

2.7 Runtime and Memory Analysis

Generic dense-transition HMM. Let T be the number of anchor sites or graph blocks and K be the number of cohort haplotype states. Computing the local target-similarity potentials requires $O(TK)$ time when each local similarity can be evaluated in constant or cached time. For a dense transition matrix, the forward recursion requires $O(K^2)$ work per position, and therefore $O(TK^2)$ total time. The backward pass computes K conditional weights per position and requires $O(TK)$ time. Hence the dense forward recursion dominates. Storing every forward vector requires $O(TK)$ memory. The current state and working vectors require an additional $O(K)$ memory.

Sparse graph compatible transitions. If the transition graph between positions $t-1$ and t has edge set E_t, then the forward pass requires $O(|E_t|)$, time at position t. The total time is $O(TK + \sum_{t=2}^{T} |E_t|)$. This form is appropriate when HMM states represent graph-valid local traversals and only a small subset of state pairs (i.e., the linkage between haploblocks) can be concatenated.

Stay–switch transitions The HMM implementation by PanMixer uses a transition model with one probability for remaining in the same donor state and another probability for switching donors [1]. Denote the transition probability

$$A_t(i,j) = \begin{cases} a_t, & i = j, \\ b_t, & i \neq j. \end{cases}$$

Let $S_{t-1} = \sum_{i=1}^{K} \alpha_{t-1}(i)$. Then $r_t(j) = \psi_t(j)[b_t S_{t-1} + (a_t - b_t)\alpha_{t-1}(j)]$. All K forward values can therefore be updated in $O(K)$ time per position. The full forward pass requires $O(TK)$ time.

Algorithm 1 HMM-based cohort-tilted τ-private path release

Require: Fixed public HMM $(\rho, A_{2:T})$, target path $p_{1:T}$, local traversals $h_{t,j}$, bounded similarities $\phi_t \in [0,1]$, weights $\beta_t \geq 0$ with $\sum_t \beta_t = 1$, and $\tau \in [0,1)$

Ensure: Sanitized graph path $\widetilde{p}$

$\eta \leftarrow \frac{1}{2} \log \frac{1+\tau}{1-\tau}$
for $t = 1, \ldots, T$ **do**
    for $j = 1, \ldots, K$ **do**
        $\psi_t(j) \leftarrow \exp(\eta \beta_t \phi_t(p_t, h_{t,j}))$
for $j = 1, \ldots, K$ **do**
    $r_1(j) \leftarrow \rho(j)\psi_1(j)$
$\alpha_1 \leftarrow r_1 / \sum_j r_1(j)$
for $t = 2, \ldots, T$ **do**
    for $j = 1, \ldots, K$ **do**
        $r_t(j) \leftarrow \psi_t(j) \sum_{i=1}^{K} \alpha_{t-1}(i) A_t(i,j)$
    $\alpha_t \leftarrow r_t / \sum_j r_t(j)$
Sample $Z_T \sim \text{Categorical}(\alpha_T)$
for $t = T-1, T-2, \ldots, 1$ **do**
    for $i = 1, \ldots, K$ **do**
        $b_t(i) \leftarrow \alpha_t(i) A_{t+1}(i, Z_{t+1})$
    $b_t \leftarrow b_t / \sum_i b_t(i)$
    Sample $Z_t \sim \text{Categorical}(b_t)$
$\widetilde{p} \leftarrow h_{1,Z_1} \| \cdots \| h_{T,Z_T}$
return $\widetilde{p}$

The backward weights for a sampled next state j are

$$b_t(i) = \begin{cases} a_{t+1}\alpha_t(j), & i = j, \\ b_{t+1}\alpha_t(i), & i \neq j. \end{cases}$$

They can also be evaluated without forming a dense $K \times K$ matrix. Consequently, the complete forward/backward sampler requires $O(TK)$ time. For a diploid cohort of n individuals, the donor-state count is at most approximately $K = 2n$ before removing duplicate haplotypes. For $n = 500$, this gives at most approximately $K = 1000$ states. The straightforward implementation stores $\alpha_1, \ldots, \alpha_T$ and uses $O(TK)$ memory.

2.8 Diploid Releases.

For a diploid target, the safest formulation treats the haplotype pair as one private input: $p = (p^{(1)}, p^{(2)})$. A joint normalized utility can be defined as $u_{\text{diploid}}(p, y) = (u(p^{(1)}, y^{(1)}) + u(p^{(2)}, y^{(2)}))/2$. Because the joint utility remains in $[0, 1]$, the same τ guarantee applies to the complete diploid release. Repeated independent releases of the same genome path should be avoided. A practical system should therefore sample one sanitized path and reuse it rather than independently resampling it for every query of the same genome.

3 Discussion

The proposed mechanism and PanMixer [1] address related but distinct privacy protection scenarios. PanMixer protects individuals contributing to the pangenome by selecting and applying block-level haplotype obfuscations. The proposed mechanism protects an external target individual's mapped path without modifying the graph. PanMixer's block and HMM machinery can nevertheless serve as the biological foundation for the target-independent path prior, while the τ-calibrated randomized sampler supplies the formal release guarantee.

The proposed mechanism supplies a worst-case pairwise guarantee: $\mathrm{TV}(Q_p^\tau, Q_q^\tau) \leq \tau$, $\forall p, q$.. This is a statement about the complete output distribution of the release mechanism, which applies even to an attacker who knows the utility function, τ and all implementation details except the private random draw. PanMixer uses different block-wise values [1]. Its HMM-based implementation computes per-block privacy scores using allele-frequency information or an HMM forward likelihood, samples replacement haplotypes, and optimizes a privacy–utility tradeoff over possible edits [1, 3]. These are valuable pangenome-specific privacy measures, but they do not by themselves provide the pairwise total variation guarantee defined here.

The HMM sampler integrates utility optimization into the probabilistic release distribution. With the stay–switch HMM structure used by PanMixer, the proposed backward sampler runs in $O(TK)$ time per haplotype.

Our implementation reuses PanMixer components while replacing its final release rule. Specifically, we use PanMixer-compatible LD blocks or graph subpaths and the cohort HMM as the target-independent prior R_D, and define ϕ_t using PanMixer utility information, weighted k-mer similarity, or mapping-seed preservation, but sample LD blocks with Algorithm 1 and release only the sampled path. This preserves PanMixer's biologically grounded cohort model while adding an explicit pairwise identifiability guarantee.

The formal guarantee applies only to the released sanitized path. It does not protect the raw genome data. The utility function should be chosen before inspecting private target-specific attack outcomes. In particular, any target-dependent pruning of the HMM state space would need a separate privacy analysis.

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